Deterministic Inventory models

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Inventory: An idle resource of an enterprise

- Fluctuation (buffer) inventory: the extra inventory a business holds to mitigate risks associated with fluctuating demand or supply chain disruptions
- Anticipation inventory: extra stock a business holds to prepare for expected increases in demand, such as during seasonal peaks or product launches
- Cycle (lot size) inventory: the average amount of inventory a business holds due to ordering or producing in batches rather than strictly meeting immediate demand.
- Transportation (pipeline) Inventory: goods that are currently being transported from one location to another within a supply chain, such as from a supplier to a warehouse or from a warehouse to a retailer.
- Decoupling inventory: The inventories needed to reduce the interdependence of various stages

$$\text{EOQ} = \sqrt{\frac{2 \times \text{Setup cost} \times \text{Annual Demand}}{\text{Annual carrying cost per unit}}}$$

- Total inventory cost T_c is given by

$$T_c = \frac{DS}{Q} + \frac{Q}{2}H$$

D: Annual demand

S: Setup or ordering cost

Q: Order quantity

H: Annual holding or carrying cost

- Differentiating T_c with respect to Q and equating it to zero,

$$\frac{dT_c}{dQ} = \frac{-DS}{Q^2} + \frac{H}{2} = 0$$

$$\Rightarrow Q = \sqrt{\frac{2DS}{H}}$$

$$T_c = DS\sqrt{\frac{H}{2DS}} + \sqrt{\frac{DS}{2H}}H$$

$$= \sqrt{2DSH}$$

- If the setup cost is $S + bQ$, then no change in EOQ

EOQ model with no shortage and several runs of unequal length

- Carrying cost

$$= \left(\frac{1}{2}Qt_1\right)H + \left(\frac{1}{2}Qt_2\right)H + \ldots\ldots + \left(\frac{1}{2}Qt_n\right)H$$

$$= \frac{1}{2}QHT$$

Multi-item Deterministic models

$$T_c = \sum_{i=1}^{n} \left(\frac{Q_i}{2} H_i + S_i \frac{D_i}{Q_i} \right)$$

Let *K* be the upper limit on total inventory cost

$$\sum_{i=1}^{n} C_{ui} Q_i \leq K$$

$$L = \sum_{i=1}^{n} \left(\frac{1}{2} H_i Q_i + \frac{S_i D_i}{Q_i} \right) + \lambda \left(\sum_{i=1}^{n} C_{ui} Q_i - K \right)$$

Marginal efficiency of capital

- Firm should invest in those alternatives that provide greater return than capital cost to borrow. The marginal efficiency of capital (MEC) is the expected rate of return on an additional unit of capital investment.

- Return on capital (Assets)

$$= \frac{\text{Profit}}{\text{Capital}}$$

$$= \left(\frac{\text{Profit}}{\text{Sales}}\right)\left(\frac{\text{Sales}}{\text{Capital}}\right)$$

$$= \text{Profit margin} \times \text{Capital turn over}$$

Dynamic Demand Models

- These models consider how dynamic demand impacts inventory management decisions, especially for products with deterioration or seasonal demand
- Increment cost model: It analyzes the additional expenses incurred from a specific business decision, like producing more units or undertaking a new project.

Dynamic demand model → helps forecast demand pattern.
Incremental cost model → helps optimize cost given the demand pattern.

A simple dynamic demand model could be:

$$D_t = a - bP_t + cD_{t-1} + \epsilon_t$$

Where:

- D_t: Demand at time t
- P_t: Price at time t
- D_{t-1}: Past demand
- ϵ_t: Random error term

👉 **Interpretation:**

- Demand today depends on **today's price** and **previous demand.**
- Positive c: Suggests trend continuation (habit-forming product).
- Negative c: Suggests stockpiling (buying now reduces future demand).

The Silver-Meal Heuristics

- Silver–Meal heuristic is a production planning method in manufacturing, composed in 1973 by Edward A. Silver and H.C. Meal.
- The Silver-Meal heuristic calculates the average cost per period by considering the setup cost and holding cost over consecutive periods. It incrementally adds future demands to the current order until the average cost per period starts increasing.

- D_t: Demand in period t
- S: Setup (ordering) cost per order
- h: Holding cost per unit per period
- n: Number of periods covered by the current order

Silver-Meal Heuristic Formula:

The **average cost per period for ordering in period t to cover demands up to period $t + n - 1$** is:

$$C(n) = \frac{S + \sum_{j=1}^{n} h\,(j-1)\,D_{t+j-1}}{n}$$

Where:

- $C(n)$: Average cost per period when covering n periods.
- S: Setup cost.
- h: Per unit per period holding cost.
- D_{t+j-1}: Demand in the $(t + j - 1)$-th period.
- $j - 1$: Number of periods a unit is held.

- Setup cost $S = 200$
- Holding cost $h = 1$ per unit per period
- Demands: $D_1 = 100$, $D_2 = 80$, $D_3 = 60$, $D_4 = 50$

Step 1: Compute for n = 1

$$C(1) = \frac{200 + 0}{1} = 200$$

Step 2: Compute for n = 2

$$C(2) = \frac{200 + 1 \times 80}{2} = \frac{280}{2} = 140$$

Step 3: Compute for n = 3

$$C(3) = \frac{200 + (1 \times 80 + 2 \times 60)}{3} = \frac{200 + 80 + 120}{3} = \frac{400}{3} \approx 133.33$$

Step 4: Compute for n = 4

$$C(4) = \frac{200 + (1 \times 80 + 2 \times 60 + 3 \times 50)}{4} = \frac{200 + 80 + 120 + 150}{4} = \frac{550}{4} = 137.5$$

- The average cost **decreases up to** $n = 3$, but increases for $n = 4$.
- So, **optimal order covers periods 1 to 3** (order 100 + 80 + 60 = 240 units at the start of period 1).

Wagner-Whitin Method

- The Wagner-Whitin algorithm is a deterministic, dynamic programming-based method used to minimize the total cost (setup and holding costs) over a finite planning horizon with time-varying demand.
- It is globally optimal, unlike heuristics like Silver-Meal which are approximate.

- D_t: Demand in period t
- S: Setup (ordering) cost per order
- h: Holding cost per unit per period
- T: Total number of periods
- $C(t)$: Minimum total cost up to period t
- $f(i, t)$: Cost of ordering in period i to cover demand up to period t

Objective:

Minimize the **total cost**:

$$\text{Total Cost} = \text{Sum of Setup Costs} + \text{Sum of Holding Costs}$$

1. **Initialization:**
   - Set $C(0) = 0$ (zero cost before starting).
2. **Cost Function:**
   - The cost of placing an order in period i to satisfy demand from period i to t is:

$$f(i,t) = S + \sum_{k=i}^{t-1} h \times (k-i) \times D_{k+1}$$

   Where:

   - $(k-i)$ is the number of periods a unit is held.

3. **Dynamic Programming Recurrence:**
   - Minimum cost up to period t is:

$$C(t) = \min_{1 \le i \le t} [C(i-1) + f(i,t)]$$

   Meaning: Try all possible previous order points and select the one that gives minimum total cost.

4. **Repeat:**
   - Continue step 3 for all periods from 1 to T.
5. **Backtracking:**
   - Determine the optimal order periods by tracing back from the final solution.

Example:

Suppose:

- Setup cost $S = 200$
- Holding cost $h = 1$ per unit per period
- Demands: $D_1 = 100, D_2 = 80, D_3 = 60$

Step 1: Calculate cost of all options

- Cost of ordering in period 1 to cover demands up to period 1:

$$f(1,1) = 200 + 0 = 200$$

- Cost of ordering in period 1 to cover demands up to period 2:

$$f(1,2) = 200 + 1 \times 80 = 280$$

- Cost of ordering in period 1 to cover demands up to period 3:

$$f(1,3) = 200 + (1 \times 80 + 2 \times 60) = 200 + 80 + 120 = 400$$

- Cost of ordering in period 2 to cover demands up to period 2:

$$f(2,2) = 200 + 0 = 200$$

- Cost of ordering in period 2 to cover demands up to period 3:

$$f(2,3) = 200 + 1 \times 60 = 260$$

- Cost of ordering in period 3 to cover demands in period 3:

$$f(3,3) = 200 + 0 = 200$$

Step 2: Dynamic programming

- $C(1) = \min\left[C(0) + f(1,1)\right] = 0 + 200 = 200$
- $C(2) = \min\left[C(0) + f(1,2), C(1) + f(2,2)\right] = \min[280, 200 + 200] = 280$
- $C(3) = \min\left[C(0) + f(1,3), C(1) + f(2,3), C(2) + f(3,3)\right]$

$$C(3) = \min[400, 200 + 260 = 460, 280 + 200 = 480] = 400$$

•Minimum total cost: 400
•Ordering points: Order in period 1 to cover all demands.

Batch size

Too large

- Inventory carrying cost more

Batch size

Too small

- Setup cost is high

- EOQ: Sensitivity analysis

$$T_c\left(Q^*\right)=\sqrt{2SHD}$$

$$T_c\left(xQ^*\right)$$

$$=\frac{1}{x}\sqrt{\frac{DSH}{2}}+x\sqrt{\frac{DSH}{2}}$$

$$=\sqrt{2SHD}\left(x+\frac{1}{x}\right)\times\frac{1}{2}$$

- Hence,

$$\frac{T_c\left(xQ^*\right)}{T_c\left(Q^*\right)}=\frac{1}{2}\left(x+\frac{1}{x}\right)$$

- Not very sensitive to Q if Q is close to Q^*
- Optimal fixed order cost equals optimal inventory cost

Types of Inventory model

- Demand: Constant, deterministic and Stochastic
- Lead times: “0”, “>0”, Stochastic
- Horizon: Single period, finite, infinite
- Products: One product, multiple products
- Capacity: Order/inventory limits, no limits
- Service: Meet all demand, shortage allowed

Profit and loss account

Expenses	Incomes
Purchase 50,000/-	Sales 60,000/-
Profit 30,000/-	Closing Inventory 20,000/-
80,000	80,000

Profit and loss account

Expenses	Incomes
Opening inventory 20,000/-	Sales 40,000/-
Purchase Nil	
Profit 20,000/-	
40000	40000